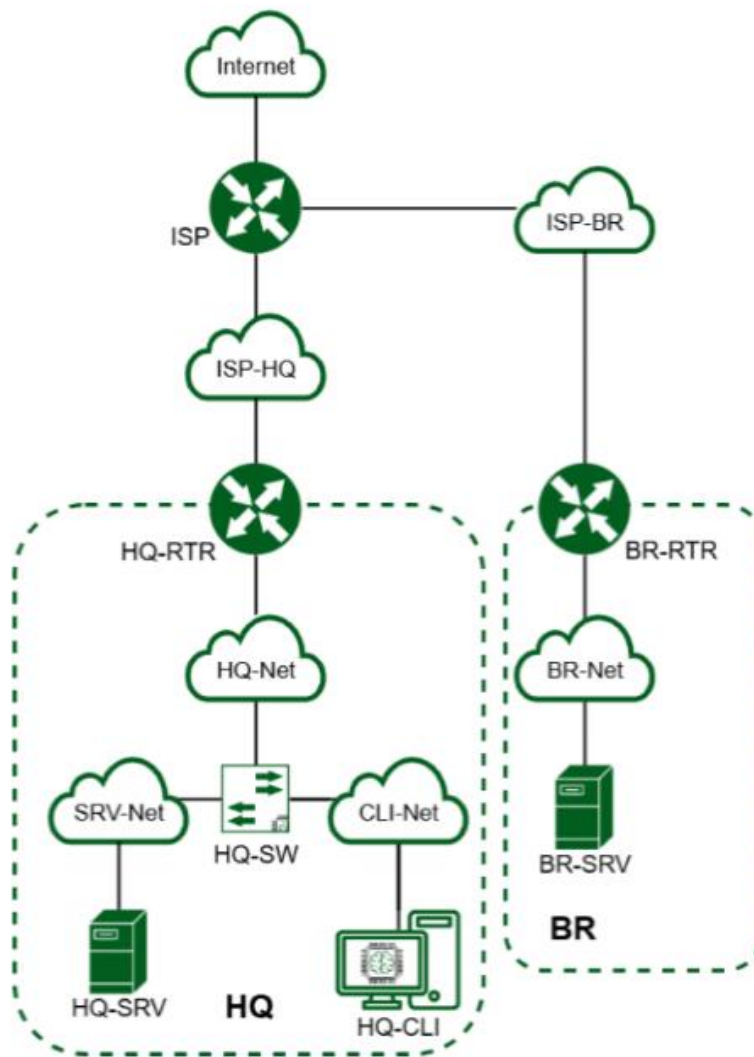


Топология сети:



Используемые ОС:

ISP, HQ-RTR, BR-RTR - Astra Linux CE

HQ-SRV, BR-SRV - Альт Сервер 10

HQ-CLI - Альт Рабочая станция 10

Таблица адресации:

Устройство	Интерфейс	IP адрес	Маска подсети	Шлюз по умолчанию
ISP	eth0	DHCP (автоматически)	-	DHCP (автоматически)
ISP	eth1	172.16.1.1/28	255.255.255.240	-
ISP	eth2	172.16.2.1/28	255.255.255.240	-
HQ-RTR	eth0	172.16.1.2/28	255.255.255.240	172.16.1.1
HQ-RTR	eth1.100	192.168.1.1/27	255.255.255.224	-
HQ-RTR	eth1.200	192.168.2.1/28	255.255.255.240	-
HQ-RTR	eth1.999	192.168.99.1/29	255.255.255.248	-
HQ-RTR	gre1 (ip tunnel)	10.0.0.1/30	255.255.255.252	-
HQ-SRV	ens18	192.168.1.2/27	255.255.255.224	192.168.1.1
HQ-CLI	ens18	DHCP (192.168.2.2/28)	255.255.255.240	DHCP (192.168.2.1)
BR-RTR	eth0	172.16.2.2/28	255.255.255.240	172.16.2.1
BR-RTR	eth1	192.168.3.1/28	255.255.255.240	-
BR-SRV	ens18	192.168.3.2/28	255.255.255.240	192.168.3.1

- команды

- проверка

Модуль 1. Настройка сетевой инфраструктуры

ISP (user P@ssw0rd):

Задание 1. Произведите базовую настройку устройств

sudo su

hostnamectl set-hostname isp.au-team.irpo; exec bash

Задание 11. Настройте часовой пояс на всех устройствах

timedatectl set-timezone Asia/Yekaterinburg

timedatectl

Задание 2. Настройте доступ к сети Интернет, на маршрутизаторе ISP

mcedit /etc/network/interfaces

```
/etc/network/interfaces [-M--] 0 L:[ 1+20 21/ 21] *(387 / 387b) <EOF>
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet dhcp

auto eth1
iface eth1 inet static
<----->address 172.16.1.1/28

auto eth2
iface eth2 inet static
<----->address 172.16.2.1/28
```

echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf

sysctl -p

iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE

iptables-save > /etc/rules.v4

crontab -e

в конец добавить:

@reboot /sbin/iptables-restore < /etc/rules.v4

@reboot /sbin/sysctl -p

echo "nameserver 8.8.8.8" > /etc/resolv.conf

systemctl restart networking

ip -br -c a

ping 8.8.8.8

HQ-RTR (user P@ssw0rd):

Задание 1. Произведите базовую настройку устройств + туннель

```
sudo su
hostnamectl set-hostname hq-rtr.au-team.irpo; exec bash
mcedit /etc/network/interfaces
```

```
GNU nano 2.7.4                                     Файл: /etc/network/interfaces

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet static
    address 172.16.1.2/28
    gateway 172.16.1.1

auto eth1
iface eth1 inet manual

auto eth1.100
iface eth1.100 inet static
    address 192.168.1.1/27
    vlan-raw-device eth1

auto eth1.200
iface eth1.200 inet static
    address 192.168.2.1/28
    vlan-raw-device eth1

auto eth1.999
iface eth1.999 inet static
    address 192.168.99.1/29
    vlan-raw-device eth1

auto gre1
iface gre1 inet tunnel
    address 10.0.0.1
    netmask 255.255.255.252
    mode gre
    local 172.16.1.2
    endpoint 172.16.2.2
    ttl 255
```

```
systemctl restart networking
```

```
ip -br -c a
ping 8.8.8.8
```

Задание 11. Настройте часовой пояс на всех устройствах

```
timedatectl set-timezone Asia/Yekaterinburg
timedatectl
```

Задание 3. Создайте локальные учетные записи на серверах HQ-SRV и BR-SRV:

```
useradd -m net_admin -u 2026
passwd net_admin
usermod -aG sudo net_admin
nano /etc/sudoers
```

```
root ALL=(ALL:ALL) ALL

# Allow members of group sudo to execute any command
%sudo ALL=(ALL:ALL) NOPASSWD: ALL

# See sudoers(5) for more information on "#include"
```

Проверка:

```
root@hq-rtr:~# su net_admin
sh-4.4$ sudo ls
sudo: unable to resolve host hq-rtr.au-team.irpo: В соединении отказано
Desktop Desktops needs SystemWallpapers Вугео Документы Загрузки Изображения Музыка Общедоступные Шаблоны
sh-4.4$ su root
Пароль:
root@hq-rtr:~#
```

Задание 8. Настройка динамической трансляции адресов маршрутизаторах HQ-RTR и BR-RTR

```
echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf
sysctl -p
iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
iptables-save > /etc/rules.v4
echo "nameserver 8.8.8.8" > /etc/resolv.conf
crontab -e
# в конец добавить:
@reboot /sbin/iptables-restore < /etc/rules.v4
@reboot /sbin/sysctl -p
systemctl restart networking
```

Задание 7. Обеспечьте динамическую маршрутизацию на маршрутизаторах HQ-RTR и BR-RTR: сети одного офиса должны быть доступны из другого офиса и наоборот

```
echo "deb [trusted=yes] https://archive.debian.org/debian buster main" >>
/etc/apt/sources.list
apt-get update && apt-get install frr
mcedit /etc/frr/daemons → ospfd=yes
systemctl restart frr
systemctl enable frr
vtysh

conf t
router ospf
network 10.0.0.0/30 area 0
network 192.168.1.0/27 area 0
network 192.168.2.0/28 area 0
network 192.168.99.0/29 area 0
```

```

exit
do wr mem
int gre1
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 P@ssw0rd
quit
do wr mem
show run
show ip ospf route
exit
systemctl restart frr
systemctl enable frr
crontab -e

```

В КОНЕЦ ДОБАВИТЬ:

```

*/2 * * * * /bin/systemctl restart frr
@reboot /sbin/iptables-restore < /etc/rules.v4
@reboot /sbin/sysctl -p

```

Задание 9. Настройте протокол динамической конфигурации хостов для сети в сторону HQ-CLI

```

apt-get install -y isc-dhcp-server
nano /etc/default/isc-dhcp-server

```

```

# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="eth1.200"
#INTERFACESv6=""

```

```

nano /etc/dhcp/dhcpd.conf

```

```
GNU nano 2.7.4                               Файл: /etc/dhcp/dhcpd.conf
# dhcpd.conf
#
# Sample configuration file for ISC dhcpd
#
# option definitions common to all supported networks
#option domain-name "example.org";
#option domain-name-servers ns1.example.org, ns2.example.org;

default-lease-time 600;
max-lease-time 7200;

# The ddns-updates-style parameter controls whether or not the server will
# attempt to do a DNS update when a lease is confirmed. We default to the
# behavior of the version 2 packages ('none', since DHCP v2 didn't
# have support for DDNS.)
ddns-update-style none;

# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
authoritative;
```

А также в конце пишем:

```
# A slightly different configuration for an internal subnet.
subnet 192.168.2.0 netmask 255.255.255.240 {
    range 192.168.2.2 192.168.2.10;
    option domain-name-servers 192.168.1.2;

    option routers 192.168.2.1;
}
```

systemctl restart isc-dhcp-server

BR-RTR: (user P@ssw0rd):

Задание 1. Произведите базовую настройку устройств + туннель

```
sudo su
hostnamectl set-hostname br-rtr.au-team.irpo; exec bash
mcedit /etc/network/interfaces
```

```
/etc/network/interfaces [----] 21 L:[ 1+24 25/ 27] *(522 / 537b) 0046 0x02E
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet static
    address 172.16.2.2/28
    gateway 172.16.2.1

auto eth1
iface eth1 inet static
    address 192.168.3.1/28

auto gre1
iface gre1 inet tunnel
    address 10.0.0.2
    netmask 255.255.255.252
    mode gre
    local 172.16.2.2
    endpoint 172.16.1.2
    ttl 255
```

```
systemctl restart networking
```

```
ip -br -c a
ping 8.8.8.8
```

Задание 11. Настройте часовой пояс на всех устройствах

```
timedatectl set-timezone Asia/Yekaterinburg
timedatectl
```

Задание 3. Создайте локальные учетные записи на серверах HQ-SRV и BR-SRV:

```
useradd -m net_admin -u 2026
passwd net_admin
usermod -aG sudo net_admin
nano /etc/sudoers
```

```
root ALL=(ALL:ALL) ALL

# Allow members of group sudo to execute any command
%sudo ALL=(ALL:ALL) NOPASSWD: ALL

# See sudoers(5) for more information on "#include"
```

Проверка:

```
root@hq-rtr:~# su net_admin
sh-4.4$ sudo ls
sudo: unable to resolve host hq-rtr.au-team.irpo: В соединении отказано
Desktop Desktops needs SystemWallpapers Вугео Документы Загрузки Изображения Музыка Общедоступные Шаблоны
sh-4.4$ su root
Пароль:
root@hq-rtr:~#
```

Задание 8. Настройка динамической трансляции адресов маршрутизаторах HQ-RTR и BR-RTR

```
echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf
sysctl -p
iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
iptables-save > /etc/rules.v4
echo "nameserver 8.8.8.8" > /etc/resolv.conf
crontab -e
# в конец добавить:
@reboot /sbin/iptables-restore < /etc/rules.v4
@reboot /sbin/sysctl -p
systemctl restart networking
```

Задание 7. Обеспечьте динамическую маршрутизацию на маршрутизаторах HQ-RTR и BR-RTR: сети одного офиса должны быть доступны из другого офиса и наоборот

```
echo "deb [trusted=yes] https://archive.debian.org/debian buster main" >>
/etc/apt/sources.list
apt-get update && apt-get install frr
mcedit /etc/frr/daemons → ospfd=yes
systemctl restart frr
systemctl enable frr
vtysh

conf t
router ospf
network 10.0.0.0/30 area 0
network 192.168.3.0/28 area 0
exit
do wr mem
int gre1
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 P@ssw0rd
quit
do wr mem
show run
show ip ospf route
exit
systemctl restart frr
```

```
systemctl enable frr  
crontab -e
```

В КОНЕЦ ДОБАВИТЬ:

```
*/2 * * * * /bin/systemctl restart frr  
@reboot /sbin/iptables-restore < /etc/rules.v4  
@reboot /sbin/sysctl -p
```

Проверка сервиса:

Возвращаемся на HQ-RTR

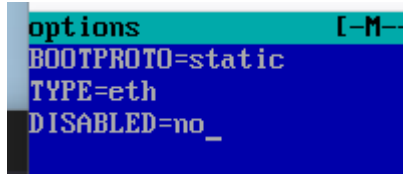
```
vtysh  
conf t  
show ip ospf neighbor  
show ip ospf route
```

```
hq-rtr.au-team.irpo(config-router)# do show ip ospf route  
===== OSPF network routing table =====  
N    10.0.0.0/30          [10] area: 0.0.0.0  
      directly attached to gre1  
N    192.168.1.0/27       [1] area: 0.0.0.0  
      directly attached to eth1.100  
N    192.168.2.0/28       [1] area: 0.0.0.0  
      directly attached to eth1.200  
N    192.168.3.0/28       [11] area: 0.0.0.0  
      via 10.0.0.2, gre1  
N    192.168.99.0/29      [1] area: 0.0.0.0  
      directly attached to eth1.999  
  
===== OSPF router routing table =====  
  
===== OSPF external routing table =====  
  
hq-rtr.au-team.irpo(config-router)#
```

HQ-SRV(root P@ssw0rd):

Задание 1. Произведите базовую настройку устройств

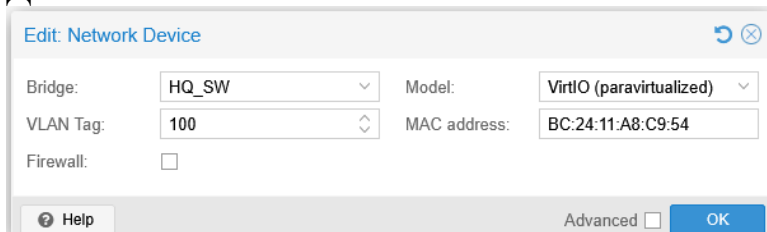
```
hostnamectl set-hostname hq-srv.au-team.irpo; exec bash  
mcedit /etc/net/ifaces/ens18/options
```



```
options  
BOOTPROTO=static  
TYPE=eth  
DISABLED=no_
```

```
echo "192.168.1.2/27" > /etc/net/ifaces/ens18/ipv4address  
echo "default via 192.168.1.1" > /etc/net/ifaces/ens18/ipv4route  
echo "nameserver 8.8.8.8" > /etc/resolv.conf  
systemctl restart networking  
ip -br -c a  
ping 8.8.8.8
```

Добавим VLAN-тег:



Edit: Network Device

Bridge: HQ_SW Model: VirtIO (paravirtualized)

VLAN Tag: 100 MAC address: BC:24:11:A8:C9:54

Firewall: ☐

Help Advanced OK

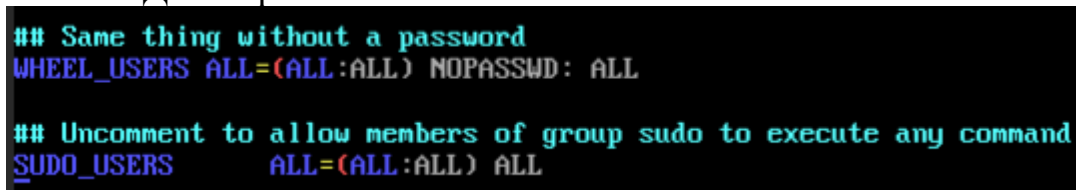
Задание 11. Настройте часовой пояс на всех устройствах

```
timedatectl set-timezone Asia/Yekaterinburg  
timedatectl
```

Задание 3. Создайте локальные учетные записи на серверах HQ-SRV и BR-SRV:

```
mcedit /etc/sudoers
```

Меняем ДВЕ строки:



```
## Same thing without a password  
WHEEL_USERS ALL=(ALL:ALL) NOPASSWD: ALL  
  
## Uncomment to allow members of group sudo to execute any command  
SUDO_USERS ALL=(ALL:ALL) ALL
```

```
sudo useradd sshuser -u 2026  
passwd sshuser  
sudo usermod -aG wheel sshuser
```

Проверка:

```

[root@br-srv ~]# su sshuser
bash: /root/.bashrc: Permission denied
bash-4.4$ sudo ls
tmp
bash-4.4$ su root
Password:
su: Authentication failure
bash-4.4$ su root
Password:
[root@br-srv ~]#

```

Задание 5. Настройте безопасный удаленный доступ на серверах HQ-SRV и BR-SRV

mcedit /etc/openssh/sshd_config

```

Port 2026
AllowUsers sshuser
PermitRootLogin no
MaxAuthTries 2
Banner /root/banner_

```

mcedit /root/banner

```

banner [-M--]
Authorized access only
-

```

systemctl restart sshd

systemctl enable sshd

Проверка:

```

[root@hq-srv ~]# ssh user@localhost -p 2026
Authorized access only
user@localhost's password:
ssh: Permission denied, please try again.
user@localhost's password:

[root@hq-srv ~]# ssh sshuser@localhost -p 2026
Authorized access only
sshuser@localhost's password:
[sshuser@hq-srv ~]#

```

Задание 10. Настройте инфраструктуру разрешения доменных имён для офисов HQ и BR

apt-get update && apt-get install dnsmasq -y

mcedit /etc/dnsmasq.conf

```

interface=*
server=8.8.8.8
#no-resolv
#no-hosts

host-record=hq-rtr.au-team.irpo,192.168.1.1
host-record=hq-srv.au-team.irpo,192.168.1.2
host-record=hq-cli.au-team.irpo,192.168.2.2

address=/br-rtr.au-team.irpo/192.168.3.1
address=/br-srv.au-team.irpo/192.168.3.2
address=/docker.au-team.irpo/172.16.1.1
address=/web.au-team.irpo/172.16.2.1

```

systemctl restart dnsmasq

systemctl enable --now dnsmasq

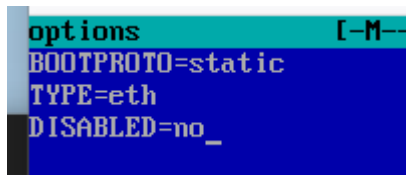
Проверка:

```
[root@hq-srv ~]# ping hq-rtr.au-team.irpo
PING hq-rtr.au-team.irpo (192.168.1.1) 56(84) bytes of data.
64 bytes from _gateway (192.168.1.1): icmp_seq=1 ttl=64 time=1.17 ms
64 bytes from _gateway (192.168.1.1): icmp_seq=2 ttl=64 time=1.10 ms
^C
--- hq-rtr.au-team.irpo ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 999ms
rtt min/avg/max/mdev = 1.095/1.134/1.174/0.039 ms
[root@hq-srv ~]#
```

BR-SRV(root P@ssw0rd):

Задание 1. Произведите базовую настройку устройств

```
hostnamectl set-hostname br-srv.au-team.irpo; exec bash  
mcedit /etc/net/ifaces/ens18/options
```



```
options  
BOOTPROTO=static  
TYPE=eth  
DISABLED=no_
```

```
echo "192.168.3.2/28" > /etc/net/ifaces/ens18/ipv4address  
echo "default via 192.168.3.1" > /etc/net/ifaces/ens18/ipv4route  
echo "nameserver 8.8.8.8" > /etc/resolv.conf  
systemctl restart networking  
ip -br -c a  
ping 8.8.8.8
```

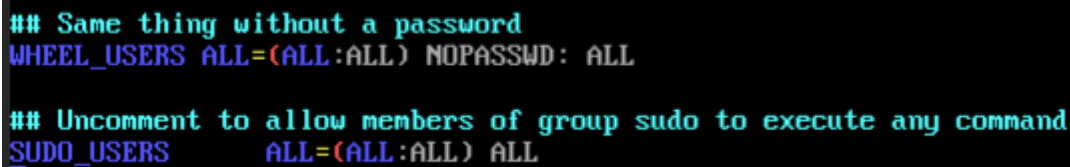
Задание 11. Настройте часовой пояс на всех устройствах

```
timedatectl set-timezone Asia/Yekaterinburg  
timedatectl
```

Задание 3. Создайте локальные учетные записи на серверах HQ-SRV и BR-SRV:

```
mcedit /etc/sudoers
```

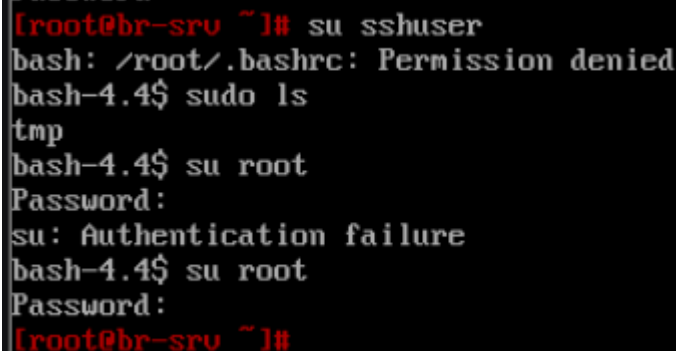
Меняем ДВЕ строки:



```
## Same thing without a password  
WHEEL_USERS ALL=(ALL:ALL) NOPASSWD: ALL  
  
## Uncomment to allow members of group sudo to execute any command  
SUDO_USERS ALL=(ALL:ALL) ALL
```

```
sudo useradd sshuser -u 2026  
passwd sshuser  
sudo usermod -aG wheel sshuser
```

Проверка:



```
[root@br-srv ~]# su sshuser  
bash: /root/.bashrc: Permission denied  
bash-4.4$ sudo ls  
tmp  
bash-4.4$ su root  
Password:  
su: Authentication failure  
bash-4.4$ su root  
Password:  
[root@br-srv ~]#
```

Задание 5. Настройте безопасный удаленный доступ на серверах HQ-SRV и BR-SRV

```
mcedit /etc/openssh/sshd_config
```

```
Port 2026
AllowUsers sshuser
PermitRootLogin no
MaxAuthTries 2
Banner /root/banner_
```

mcedit /root/banner

```
banner [-M--]
Authorized access only
_
```

systemctl restart sshd

systemctl enable sshd

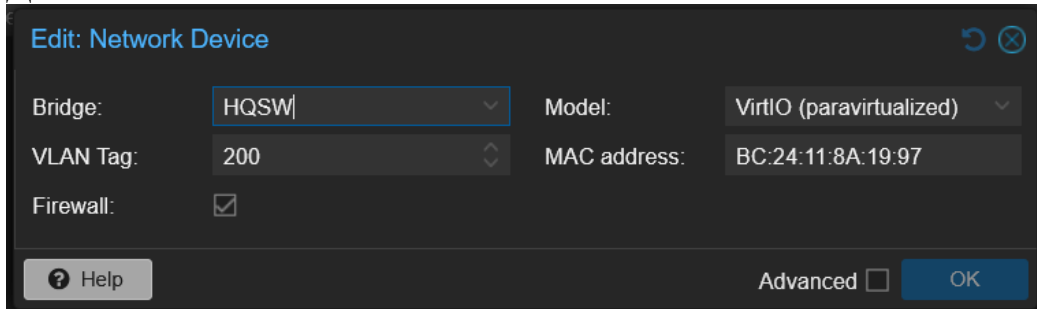
Проверка:

```
(root@hq-srv ~)# ssh user@localhost -p 2026
Authorized access only
user@localhost's password:
ssh: Permission denied, please try again.
user@localhost's password:

(root@hq-srv ~)# ssh sshuser@localhost -p 2026
Authorized access only
sshuser@localhost's password:
[sshuser@hq-srv ~]$
```

HQ-CLI(user P@ssw0rd):

Добавим VLAN-тег:



Edit: Network Device

Bridge: HQSWM Model: VirtIO (paravirtualized)

VLAN Tag: 200 MAC address: BC:24:11:8A:19:97

Firewall: ☒

Help Advanced ☐ OK

Открываем терминал.

su

Вводим пароль: P@ssw0rd

Задание 1. Произведите базовую настройку устройств

hostnamectl set-hostname hq-cli.au-team.irpo; exec bash

Задание 11. Настройте часовой пояс на всех устройствах

timedatectl set-timezone Asia/Yekaterinburg

timedatectl

Модуль 2. Настройка сетевой инфраструктуры

ISP (user P@ssw0rd):

Задание 4. Настройте службу сетевого времени на базе сервиса chrony на маршрутизаторе ISP:

```
apt-get update && apt-get install chrony -y
nano/etc/chrony/chrony.conf
```

Комментируем старый ntp:

```
#Welcome to the chrony configuration file. See chrony.conf(5) for more
# information about usable directives.
#pool 2.debian.pool.ntp.org iburst
```

И в конец файла пишем:

```
server 0.ru.pool.ntp.org iburst prefer minstratum 4
local stratum 5
allow 0.0.0.0/0
```

```
systemctl restart chronyd
```

Проверка:

```
root@isp:/home/igor# chronyc tracking
Reference ID      : 7F7F0101 ()
Stratum          : 5
Ref time (UTC)   : Sat Nov 22 09:37:37 2025
System time      : 0.000000000 seconds slow of NTP time
Last offset      : +0.000000000 seconds
RMS offset       : 0.000000000 seconds
Frequency        : 20.809 ppm slow
Residual freq    : +0.000 ppm
Skew             : 0.000 ppm
Root delay       : 0.000000000 seconds
Root dispersion  : 0.000000000 seconds
Update interval  : 0.0 seconds
Leap status      : Normal
root@isp:/home/igor# chronyc sources
210 Number of sources = 1
MS Name/IP address      Stratum Poll Reach LastRx Last sample
-----
^* mskm9-ntp02c.ntppool.yan> 4 6 17 0 -73us[-1393us] +/- 17ms
```

Задание 9. Настройте веб-сервер nginx как обратный прокси-сервер на ISP +

Задание 10. На маршрутизаторе ISP настройте web-based аутентификацию

```
apt-get install ssh nginx apache2-utils -y
htpasswd -bc /etc/nginx/.htpasswd WEB 'P@ssw0rd'
chmod 644 /etc/nginx/.htpasswd
rm -rf /etc/nginx/sites-available/default
mcedit /etc/nginx/sites-available/default
```

```

/etc/nginx/sites-available/default [----] 0 L:[ 1+27 28/ 28] *(669 / 669b) <EOF>
server {
<----->listen 80;
<----->server_name web.au-team.irpo;
<----->
<----->location / {
<-----><----->proxy_pass http://172.16.1.2:8080;
<-----><----->proxy_set_header Host $host;
<-----><----->proxy_set_header X-Real-IP $remote_addr;
<-----><----->proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
<-----><----->proxy_set_header X-Forwarded-Proto $scheme;
<-----><----->auth_basic 'Web-Authorization';
<-----><----->auth_basic_user_file /etc/nginx/.htpasswd;
<----->}
}

server {
<----->listen 80;
<----->server_name docker.au-team.irpo;
<----->
<----->location / {
<-----><----->proxy_pass http://172.16.2.2:8080;
<-----><----->proxy_set_header Host $host;
<-----><----->proxy_set_header X-Real-IP $remote_addr;
<-----><----->proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
<-----><----->proxy_set_header X-Forwarded-Proto $scheme;
<----->}
}

```

ИЛИ

cd /etc/nginx/sites-available/

wget <http://file.danildebil.ru:1234/default>

nginx -t

systemctl restart nginx

HQ-RTR (user P@ssw0rd):

Задание 8. На маршрутизаторах сконфигурируйте статическую трансляцию портов

```
apt-get update && apt-get install ssh
iptables -t nat -A PREROUTING -d 172.16.1.2 -p tcp --dport 8080 -j DNAT --to-destination 192.168.1.2:80
iptables -t nat -A PREROUTING -d 172.16.1.2 -p tcp --dport 2026 -j DNAT --to-destination 192.168.1.2:2026
iptables-save > /etc/rules.v4
```

BR-RTR: (user P@ssw0rd):

Задание 4. Настройте службу сетевого времени на базе сервиса chrony на маршрутизаторе ISP:

```
apt-get update && apt-get install ssh chrony -y  
mcedit /etc/chrony/chrony.conf
```

Комментируем следующие строки:

```
GNU nano 2.7.4                                Файл: /etc/chrony/chrony.conf  
# Welcome to the chrony configuration file. See chrony.conf(5) for more  
# information about usable directives.  
#pool 2.debian.pool.ntp.org iburst  
  
# This directive specify the location of the file containing ID/key pairs for  
# NTP authentication.  
keyfile /etc/chrony/chrony.keys  
  
# This directive specify the file into which chronyd will store the rate  
# information.  
driftfile /var/lib/chrony/chrony.drift  
  
# Uncomment the following line to turn logging on.  
#log tracking measurements statistics  
  
# Log files location.  
logdir /var/log/chrony  
  
# Stop bad estimates upsetting machine clock.  
maxupdateskew 100.0  
  
# This directive enables kernel synchronisation (every 11 minutes) of the  
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.  
#rtcsync
```

В конце прописываем:

```
server 172.16.1.1 iburst
```

```
systemctl restart chrony  
systemctl enable chrony
```

Задание 8. На маршрутизаторах сконфигурируйте статическую трансляцию портов

```
apt-get update && apt-get install ssh  
iptables -t nat -A PREROUTING -d 172.16.2.2 -p tcp --dport 8080 -j DNAT --to-  
destination 192.168.3.2:8080  
iptables -t nat -A PREROUTING -d 172.16.2.2 -p tcp --dport 2026 -j DNAT --to-  
destination 192.168.3.2:2026  
iptables-save > /etc/rules.v4
```

HQ-SRV(root P@ssw0rd):

Задание 10. Настройте инфраструктуру разрешения доменных имён для офисов HQ и BR

```
mcedit /etc/dnsmasq.conf
```

Добавите строку над `server = 8.8.8.8`:

```
server=/au-team.irpo/192.168.3.2  
server=8.8.8.8
```

Задание 4. Настройте службу сетевого времени на базе сервиса chrony на маршрутизаторе ISP

```
apt-get update && apt-get install ssh chrony -y  
mcedit /etc/chrony/chrony.conf
```

Комментируем следующие строки:

```
GNU nano 2.7.4                                Файл: /etc/chrony/chrony.conf  
# Welcome to the chrony configuration file. See chrony.conf(5) for more  
# information about usable directives.  
#pool 2.debian.pool.ntp.org iburst  
  
# This directive specify the location of the file containing ID/key pairs for  
# NTP authentication.  
keyfile /etc/chrony/chrony.keys  
  
# This directive specify the file into which chronyd will store the rate  
# information.  
driftfile /var/lib/chrony/chrony.drift  
  
# Uncomment the following line to turn logging on.  
#log tracking measurements statistics  
  
# Log files location.  
logdir /var/log/chrony  
  
# Stop bad estimates upsetting machine clock.  
maxupdateskew 100.0  
  
# This directive enables kernel synchronisation (every 11 minutes) of the  
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.  
#rtcsync
```

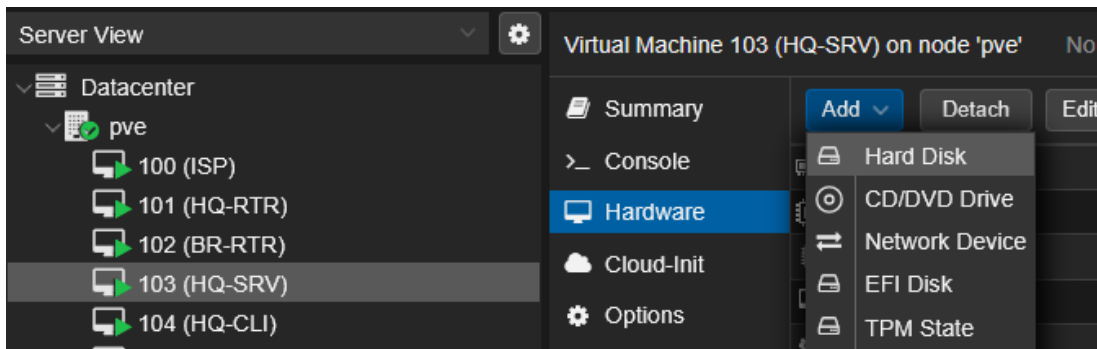
В конце прописываем:

```
server 172.16.1.1 iburst
```

```
systemctl restart chrony  
systemctl enable chrony
```

Задание 2. Сконфигурируйте файловое хранилище на сервере HQ-SRV

Для начала нужно подключить пару дисков по 1Гб к машине для дальнейшего их объединения в RAID 0.



lsblk

```
(root@hq-srv ~)# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   32G  0 disk
├─sda1       8:1    0   977M  0 part [SWAP]
└─sda2       8:2    0   31G  0 part /
sdb          8:16    0    1G  0 disk
sdc          8:32    0    1G  0 disk
sr0         11:0    1   5.1G  0 rom
```

```
/sbin/mdadm --create /dev/md0 --level=0 --raid-device=2 /dev/sdb /dev/sdc
```

```
(root@hq-srv ~)# mdadm --create /dev/md0 --level=0 --raid-device=2 /dev/sdb /dev/sdc
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
(root@hq-srv ~)#
```

```
/sbin/mdadm --detail --scan > /etc/mdadm.conf
```

```
(root@hq-srv ~)# cat /etc/mdadm.conf
ARRAY /dev/md0 metadata=1.2 name=hq-srv.au-team.irpo:0 UUID=b144da30:cf492519:a3f6651b:99b69c83
```

```
fdisk /dev/md0
```

```
(root@hq-srv ~)# fdisk /dev/md0

Welcome to fdisk (util-linux 2.39.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x1a7e26ab.

Command (m for help): _
```

1. n
2. p
- 3,4,5. скип
6. w

lsblk /dev/md0

```
(root@hq-srv ~)# lsblk /dev/md0
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
md0          9:0    0    2G  0 raid0
└─md0p1     259:0    0    2G  0 part
```

```
mkfs.ext4 /dev/md0p1
```

```
[root@hq-srv ~]# mkfs.ext4 /dev/md0p1
mke2fs 1.46.2 (28-Feb-2021)
Discarding device blocks: done
Creating filesystem with 523808 4k blocks and 130816 inodes
Filesystem UUID: 5e0e2b26-0d23-451d-9197-b2fcc49f80f8
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

mkdir /raid

mount /dev/md0p1 /raid

```
[root@hq-srv ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sda          8:0    0   32G  0 disk
├─sda1       8:1    0   977M  0 part  [SWAP]
└─sda2       8:2    0   31G  0 part  /
sdb          8:16   0    1G  0 disk
└─md0        9:0    0    2G  0 raid0
   └─md0p1   259:0   0    2G  0 part  /raid
sdc          8:32   0    1G  0 disk
└─md0        9:0    0    2G  0 raid0
   └─md0p1   259:0   0    2G  0 part  /raid
sr0         11:0    1   5.1G  0 rom
```

export EDITOR=mcedit

systemctl --force --full edit raid.mount

```
[Unit]
Description=Mount RAID0

[Mount]
What=/dev/md0p1
Where=/raid
Type=ext4
Options=defaults

[Install]
WantedBy=multi-user.target
```

systemctl enable raid.mount --now

mount -a

df -h

```
[root@hq-srv ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
udevfs          5.0M  100K  5.0M   2% /dev
runfs           2.0G  624K  2.0G   1% /run
/dev/sda2       31G   3.2G   26G  11% /
tmpfs           2.0G   4.0K  2.0G   1% /dev/shm
tmpfs           2.0G     0  2.0G   0% /tmp
tmpfs          392M     0  392M   0% /run/user/0
/dev/md0p1      2.0G   24K  1.9G   1% /raid
```

Задание 3. Настройте сервер сетевой файловой системы (nfs) на HQ-SRV

```
apt-get install nfs-server -y
mkdir -p /raid/nfs
chmod 777 /raid/nfs
mcedit /etc/exports
```

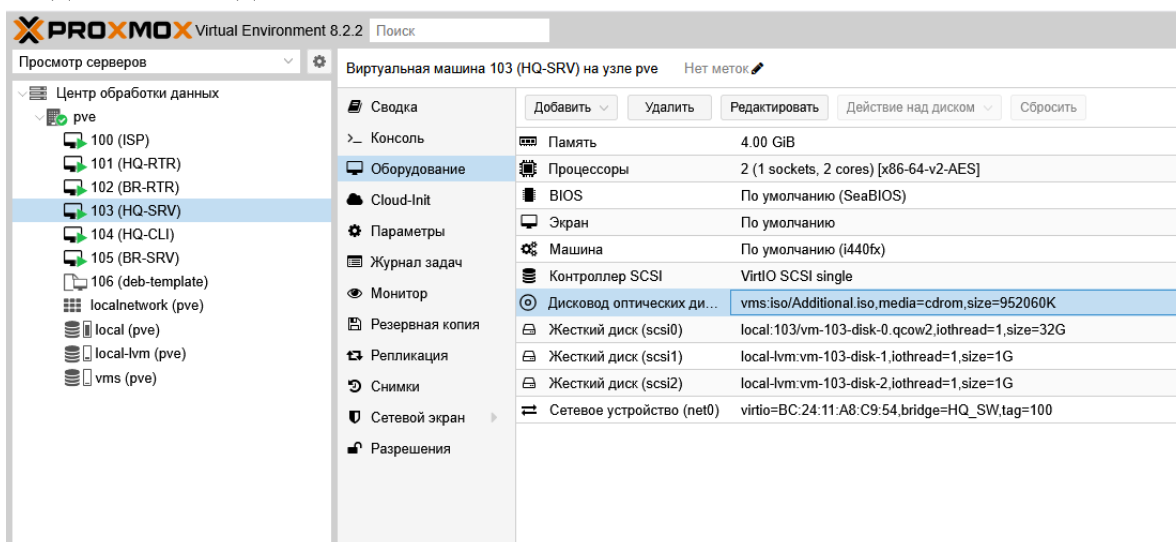
```
exports      [BM--] 50 L:[ 1+ 0  1/ 4] *(50
/srv/public -ro,insecure,no_subtree_check,fsid=1 *
/raid/nfs<----->192.168.2.0/28(rw,no_subtree_check)
#/srv/share -rw,insecure,fsid=0,sec=krb5 *
```

```
exportfs -a
systemctl restart nfs-server
systemctl enable --now nfs-server
```

Задание 7. Разверните веб приложение на сервере HQ-SRV

```
apt-get install lamp-server -y
```

Подключение диска Additional.iso:



```
mount /dev/sr0 /mnt/
```

```
ls /mnt/
```

```
(root@hq-srv ~)# ls /mnt
Users.csv  docker  playbook  web
(root@hq-srv ~)#
```

```
cp /mnt/web/index.php /var/www/html
cp /mnt/web/logo.png /var/www/html
mcedit /var/www/html/index.php
```

```
index.php      [-M--] 14
<?php
$servername = "localhost";
$username   = "web";
$password   = "P@ssw0rd";
$dbname     = "webdb";
```

```
mcedit /etc/httpd2/conf/sites-available/default.conf
```

```

default.conf  [-H-]  9 L:1 34+35 69/149] *(2196/5014b) 0060 0x03C
>#
>#ServerName www.example.com:80
>#
># DocumentRoot: The directory out of which you will serve your
># documents. By default, all requests are taken from this directory, but
># symbolic links and aliases may be used to point to other locations.
>#
DocumentRoot "/var/www/html"
>#
># Each directory to which Apache has access can be configured with respect
># to which services and features are allowed and/or disabled in that
># directory (and its subdirectories).
>#
># First, we configure the "default" to be a very restrictive set of
># features.
>#
<Directory />
<-----># Summary: Configure for all dirs by default
<-----># Requires: Directory-root
<----->Include conf/include/Directory_root_default.conf
</Directory>
>#
># Note that from this point forward you must specifically allow
># particular features to be enabled - so if something's not working as
># you might expect, make sure that you have specifically enabled it
># below.
>#
>#
># This should be changed to whatever you set DocumentRoot to.
>#
<Directory "/usr/share/doc/indexhtml/">
<Directory "/var/www/html/">
<-----># Summary: Configure for html documents in DocumentRoot
<-----># Requires: Directory-html
<----->Include conf/include/Directory_html_default.conf
</Directory>

```

systemctl enable --now mariadb

mariadb -u root

CREATE DATABASE webdb;

CREATE USER 'web'@'localhost' IDENTIFIED BY 'P@ssw0rd';

GRANT ALL PRIVILEGES ON webdb.* TO 'web'@'localhost' WITH GRANT OPTION;

Exit

mariadb -u web -p -D webdb < /mnt/web/dump.sql

Проверка:

```

[root@hq-srv ~]# mariadb
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 34
Server version: 10.11.14-MariaDB-alt1 (ALT p11)

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

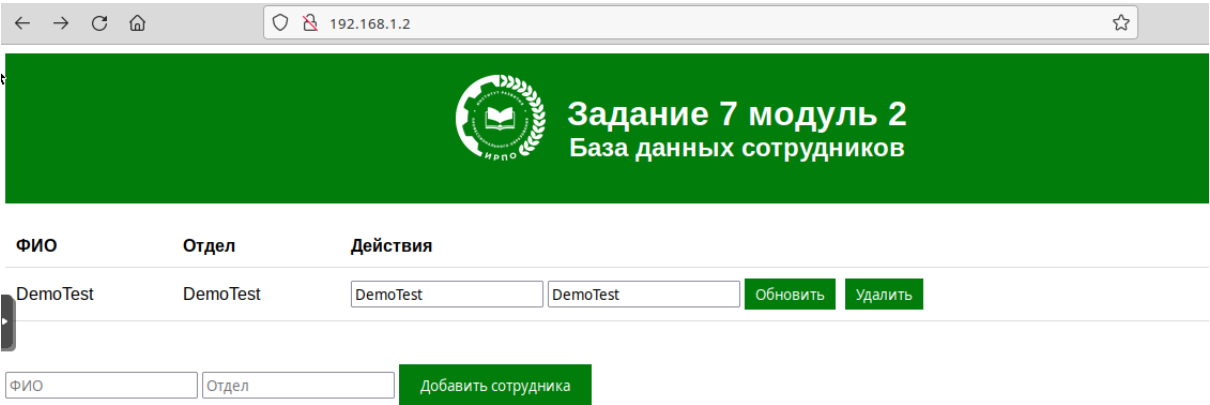
MariaDB [(none)]> USE webdb;
Database changed
MariaDB [webdb]> SHOW TABLES;
+-----+
| Tables_in_webdb |
+-----+
| employees        |
+-----+
1 row in set (0.000 sec)

MariaDB [webdb]> _

```

```
systemctl restart httpd2
systemctl enable --now httpd2
```

Проверка:



BR-SRV(root P@ssw0rd):

Задание 4. Настройте службу сетевого времени на базе сервиса chrony на маршрутизаторе ISP:

```
apt-get update && apt-get install ssh chrony -y  
mcedit /etc/chrony/chrony.conf
```

Комментируем следующие строки:

```
GNU nano 2.7.4                                Файл: /etc/chrony/chrony.conf  
  
# Welcome to the chrony configuration file. See chrony.conf(5) for more  
# information about usable directives.  
#pool 2.debian.pool.ntp.org iburst  
  
# This directive specify the location of the file containing ID/key pairs for  
# NTP authentication.  
keyfile /etc/chrony/chrony.keys  
  
# This directive specify the file into which chronyd will store the rate  
# information.  
driftfile /var/lib/chrony/chrony.drift  
  
# Uncomment the following line to turn logging on.  
#log tracking measurements statistics  
  
# Log files location.  
logdir /var/log/chrony  
  
# Stop bad estimates upsetting machine clock.  
maxupdateskew 100.0  
  
# This directive enables kernel synchronisation (every 11 minutes) of the  
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.  
#rtcsync
```

В конце прописываем:

```
server 172.16.1.1 iburst
```

```
systemctl restart chrony  
systemctl enable chrony
```

Задание 1. Настройте контроллер домена Samba DC на сервере BR-SRV

```
apt-get update && apt-get install task-samba-dc -y  
rm -rf /etc/samba/smb.conf  
samba-tool domain provision  
    Realm: AU-TEAM.IRPO  
    Domain: AU-TEAM  
    Server Role: dc  
    DNS backend: SAMBA_INTERNAL  
    DNS forwarder IP address: 192.168.1.2
```

```
cp /var/lib/samba/private/krb5.conf /etc/krb5.conf
systemctl enable samba --now
```

```
mcedit /etc/samba/smb.conf
```

Добавить:

```
interfaces = lo ens18
```

```
bind interfaces only = yes
```

```
# Global parameters
[global]
<----->dns forwarder = 192.168.1.2
<----->netbios name = BR-SRV
<----->realm = AU-TEAM.IRPO
<----->server role = active directory domain controller
<----->workgroup = AU-TEAM
<----->interfaces = lo ens18
<----->bind interfaces only = yes_

[sysvol]
<----->path = /var/lib/samba/sysvol
<----->read only = No

[netlogon]
<----->path = /var/lib/samba/sysvol/au-team.irpo/scripts
<----->read only = No
```

```
systemctl restart samba
```

```
systemctl enable --now samba
```

Проверка:

```
samba-tool domain info 127.0.0.1
```

```
samba-tool domain info 192.168.3.2
```

```
[root@br-srv ~]# samba-tool domain info 127.0.0.1
Forest           : au-team.irpo
Domain           : au-team.irpo
Netbios domain   : AU-TEAM
DC name          : br-srv.au-team.irpo
DC netbios name  : BR-SRV
Server site      : Default-First-Site-Name
Client site      : Default-First-Site-Name
```

```
samba-tool user create hquser1 P@ssw0rd
```

```
samba-tool user create hquser2 P@ssw0rd
```

```
samba-tool user create hquser3 P@ssw0rd
```

```
samba-tool user create hquser4 P@ssw0rd
```

```
samba-tool user create hquser5 P@ssw0rd
```

```
samba-tool user list
```

```
samba-tool group add hq
```

```
samba-tool group addmembers hq hquser1
```

```
samba-tool group addmembers hq hquser2
```

```
samba-tool group addmembers hq hquser3
```

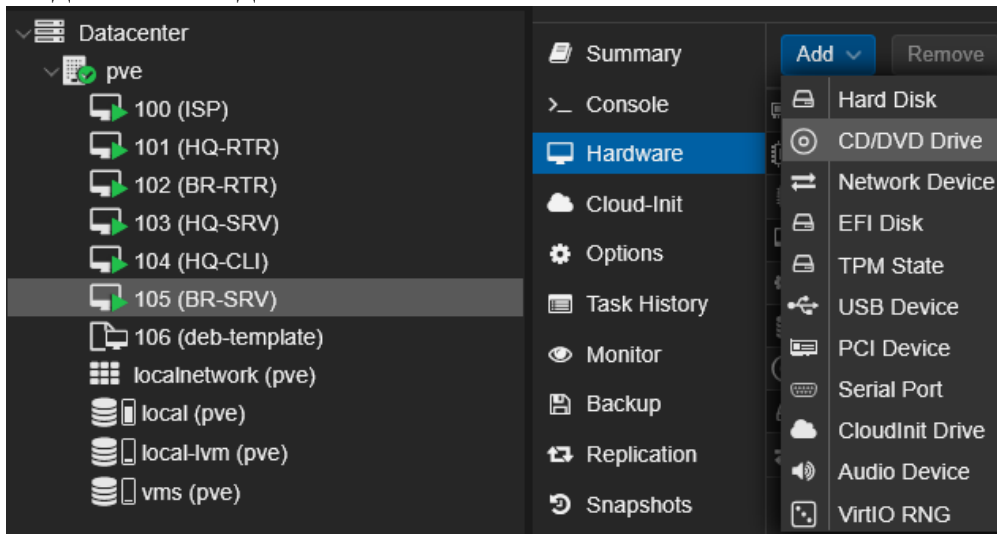
```
samba-tool group addmembers hq hquser4
```

```
samba-tool group addmembers hq hquser5
```

Задание 6. Разверните веб приложение в docker на сервере BR-SRV

```
apt-get install docker-compose docker-engine -y
systemctl enable --now docker
systemctl status docker
```

Подключение диска Additional.iso:



```
mount /dev/sr0 /mnt/
mkdir -p /root/docker/
cp -r /mnt/docker /root/
docker image load -i /root/docker/site_latest.tar
docker image load -i /root/docker/mariadb_latest.tar
mkdir -p /root/testapp/
mcedit /root/testapp/docker-compose.yaml
```

```
docker-compose.yaml  [-M--]  0 L:[ 1+26 27/ 27] *(611 / 611b) <EOF>
services:
  testapp:
    image: site:latest
    container_name: testapp
    restart: always
    depends_on:
      - db
    ports:
      - 8080:8080
    environment:
      DB_TYPE: maria
      DB_HOST: db
      DB_NAME: testdb
      DB_PORT: 3306
      DB_USER: test
      DB_PASS: P@ssw0rd
  db:
    image: mariadb:10.11
    container_name: db
    restart: always
    environment:
      MARIADB_DATABASE: testdb
      MARIADB_USER: test
      MARIADB_PASSWORD: P@ssw0rd
      MARIADB_ROOT_PASSWORD: P@ssw0rd
```

ИЛИ

```
apt-get install wget -y
cd /root/testapp/
wget http://file.danildebil.ru:1234/docker-compose.yaml
```

docker compose up -d

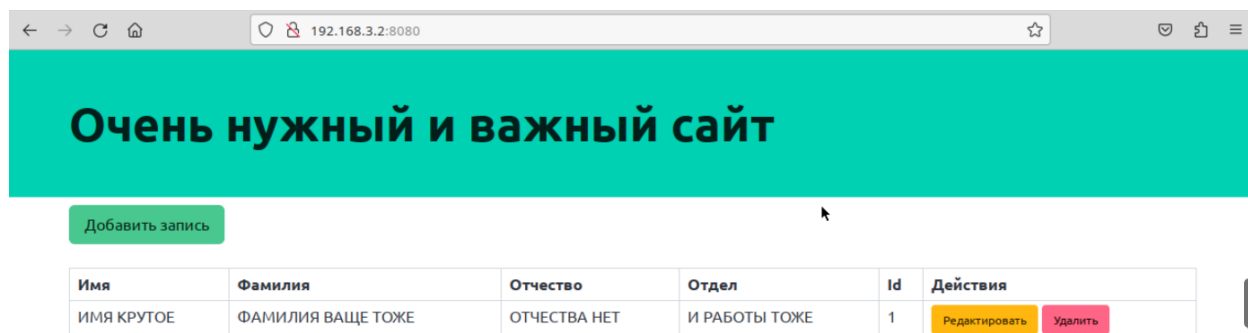
```
[root@br-srv ~]# docker compose up -d
[+] Running 2/2
 ? Container db      Running
 ? Container testapp Started
[root@br-srv ~]#
```

docker ps

```
[root@br-srv ~]# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
8e3e6a7546b5   site:latest    "sh -c 'python3 -m ap"   About a minute ago Up About a minute 0.0.0.0:8080->8080/tcp, :::8080->8080/tcp testapp
31d8de0ee451   postgres:15-alpine "docker-entrypoint.sh"    About a minute ago Up About a minute 5432/tcp db
```

Проверка:

На HQ-CLI <http://192.168.3.2:8080>



Задание 5. Сконфигурируйте ansible на сервере BR-SRV

```
apt-get install ansible sshpass -y
mcedit /etc/ansible/hosts
```

```
hosts      [-----] 45 L:1  1+ 4  5/  51 *(34b / 392b) 0117 0x075
[clients]
hq-rtr ansible_host=192.168.1.1 ansible_user=net_admin ansible_port=22 ansible_password=P0ssw0rd
br-rtr ansible_host=192.168.3.1 ansible_user=net_admin ansible_port=22 ansible_password=P0ssw0rd
hq-srv ansible_host=192.168.1.2 ansible_user=sshuser ansible_port=2026 ansible_password=P0ssw0rd
hq-cli ansible_host=192.168.2.2 ansible_user=user ansible_port=22 ansible_password=P0ssw0rd
```

```
rm -rf /etc/ansible/ansible.cfg
mcedit /etc/ansible/ansible.cfg
```

```
ansible.cfg  [-M--] 30 L:1  1+ 3  4/  41 *(96 / 96b) <EOF>
[defaults]
interpreter_python = python3
host_key_checking = False
inventory = /etc/ansible/hosts
```

Проверка:

ansible all -m ping

```

[root@br-srv ~]# ansible all -m ping
HQ-CLI | UNREACHABLE! => {
  "changed": false,
  "msg": "Failed to connect to the host via ssh: ssh: connect to host 192.168.2.2 port 22: Connection refused",
  "unreachable": true
}
BR-RTR | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
HQ-RTR | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
HQ-SRV | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
[root@br-srv ~]#

```

Ашибочка

На HQ-CLI:

```

apt-get update && apt-get install openssh
systemctl enable --now sshd

```

Возвращаемся на BR-SRV:

```

[root@br-srv ~]# ansible all -m ping
BR-RTR | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
HQ-RTR | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
HQ-SRV | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
HQ-CLI | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
[root@br-srv ~]#

```

HQ-CLI(user P@ssw0rd):

Задание 4. Настройте службу сетевого времени на базе сервиса chrony на маршрутизаторе ISP

```
apt-get update && apt-get install ssh chrony -y  
mcedit /etc/chrony/chrony.conf
```

Комментируем следующие строки:

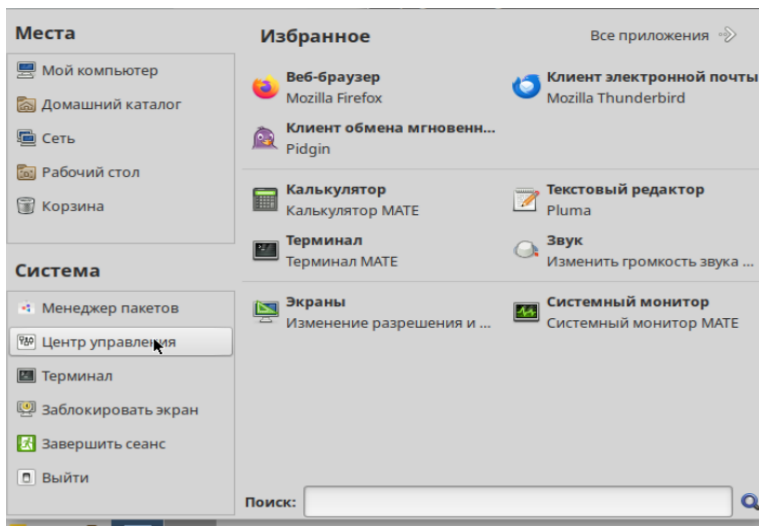
```
GNU nano 2.7.4                                Файл: /etc/chrony/chrony.conf  
  
# Welcome to the chrony configuration file. See chrony.conf(5) for more  
# information about usable directives.  
#pool 2.debian.pool.ntp.org iburst  
  
# This directive specify the location of the file containing ID/key pairs for  
# NTP authentication.  
keyfile /etc/chrony/chrony.keys  
  
# This directive specify the file into which chronyd will store the rate  
# information.  
driftfile /var/lib/chrony/chrony.drift  
  
# Uncomment the following line to turn logging on.  
#log tracking measurements statistics  
  
# Log files location.  
logdir /var/log/chrony  
  
# Stop bad estimates upsetting machine clock.  
maxupdateskew 100.0  
  
# This directive enables kernel synchronisation (every 11 minutes) of the  
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.  
#rtcsync
```

В конце прописываем:

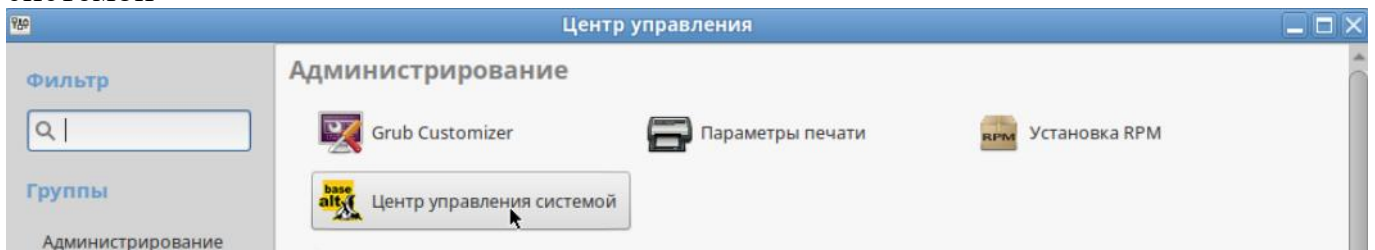
```
server 172.16.1.1 iburst
```

```
systemctl restart chrony  
systemctl enable chrony
```

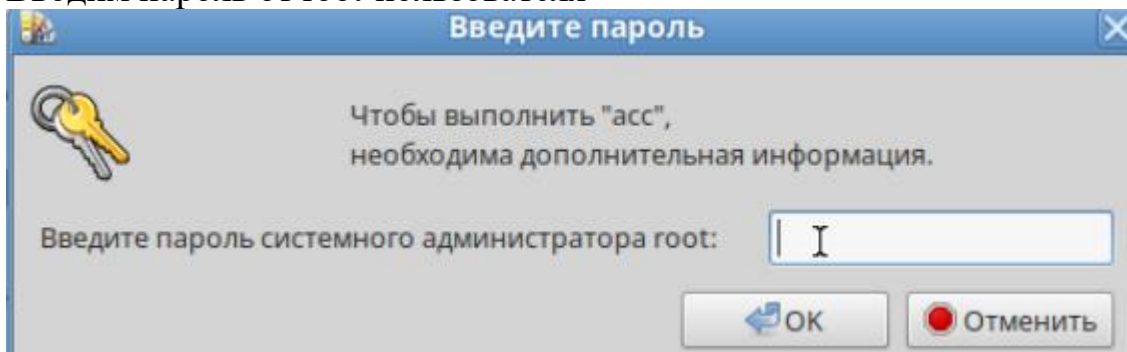
Задание 1. Настройте контроллер домена Samba DC на сервере BR-SRV (вход в домен)



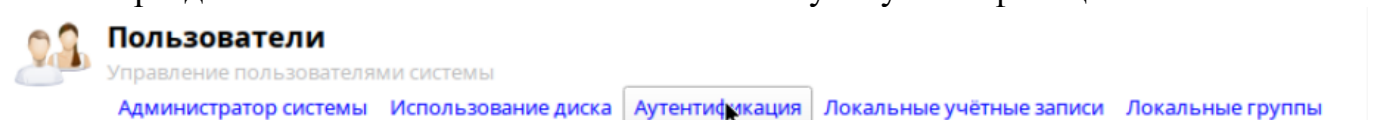
Далее в категории “Администрирование” кликаем по иконке “Центр управления системой”



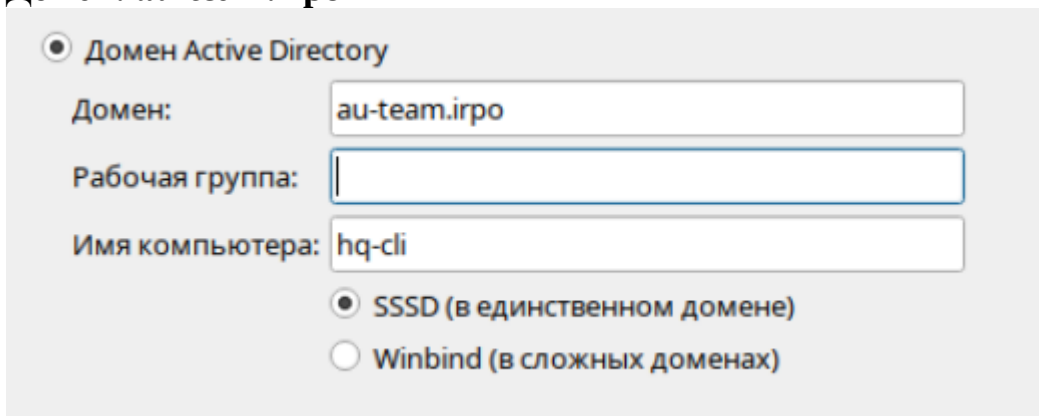
Вводим пароль от root-пользователя



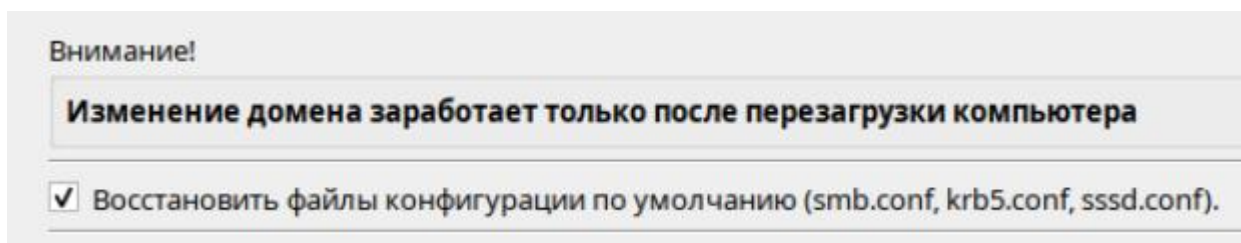
Затем в разделе “Пользователи” кликаем на иконку “Аутентификация”



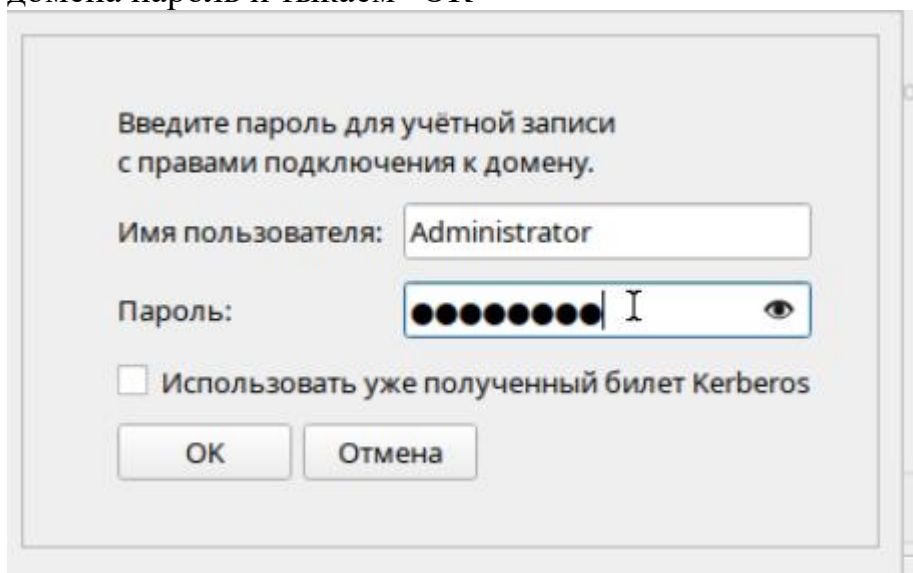
После выбираем пункт “Домен Active Directory” и заполняем поле домена
Домен: au-team.irpo



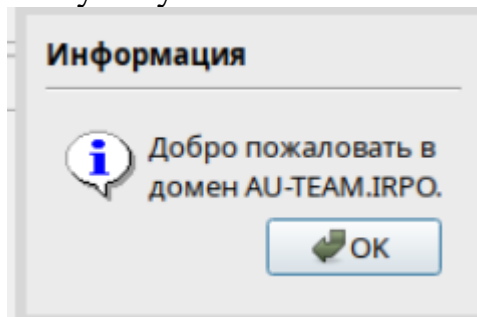
Снизу ставим флажок “Восстановить файлы конфигурации по умолчанию”



Далее в появившемся окне вводим созданный при разворачивании контроллера домена пароль и нажимаем “ОК”



В случае успеха появится вот-такое окно!



```
nano /etc/sudoers
```

```
#User_Alias      HQUSERS = hquser1, hquser2, hquser3, hquser4,
%hq      ALL = NOPASSWD: /usr/bin/cat, /bin/grep, /usr/bin/id
#HQUSERS      ALL=(ALL:ALL)  ALL
##
```

```
chmod 4755 /usr/bin/sudo
```

Задание 3. Настройте сервер сетевой файловой системы (nfs) на HQ-SRV

```
apt-get install nfs-clients -y
```

```
mkdir /mnt/nfs
```

```
echo "HQ-SRV:/raid/nfs /mnt/nfs nfs defaults,_netdev 0 0" >> /etc/fstab
```

↑ тут нижнее подчеркивание

```
mount -a
```

```
df -h
```

```
[root@hq-CLI ~]# df -h
Файловая система    Размер  Использовано  Дост  Использовано%  Смонтировано в
udevfs               5,0M      100K          5,0M           2% /dev
runfs                984M      1,3M        983M           1% /run
/dev/sda2            30G       9,0G        20G           32% /
tmpfs                984M           0        984M           0% /dev/shm
tmpfs                984M       12K        984M           1% /tmp
tmpfs               197M       72K        197M           1% /run/user/90201104
tmpfs               197M       76K        197M           1% /run/user/500
tmpfs               197M       88K        197M           1% /run/user/90201107
192.168.1.2:/raid/nfs 2,0G           0         1,9G           0% /mnt/nfs
```

Задание 11. Удобным способом установите приложение Яндекс Браузер на HQ-CLI

```
apt-get install yandex-browser-stable -y
```

Конечная проверка:

The screenshot shows a web browser window with the address bar displaying 'web.au-team.irpo/'. The page has a green header with a logo and the text 'Задание 7 модуль 2 База данных сотрудников'. Below the header is a table with columns 'ФИО', 'Отдел', and 'Действия'. The table contains one row with the value 'DemoTest' in the first two columns. To the right of the table are two buttons: 'Обновить' and 'Удалить'. Below the table is a form with two input fields labeled 'ФИО' and 'Отдел', and a green button labeled 'Добавить сотрудника'.

ФИО	Отдел	Действия
DemoTest	DemoTest	DemoTest DemoTest Обновить Удалить

